



Lecture 6

Robustness to Effect Heterogeneity

Dave Armstrong

Today's Robustness Question

What assumption/abstraction are we perturbing?

- That the nature of the relationship between x and y regarding moderation is appropriately specified.

Why might this assumption fail?

- The estimated relationship is unconditional, but the effect varies across cases.
- The estimated relationship is moderated, but the unconditional relationship is also plausible.
- The estimated relationship is moderated, but the form of moderation is misspecified (e.g., wrong moderator, omitted moderator, or incorrect functional form).

Today's Robustness Question

What assumption/abstraction are we perturbing?

- That the nature of the relationship between x and y regarding moderation is appropriately specified.

What kinds of inferential instability can result?

- Misleading conclusions about the average effect.
- Improper estimation of effect sizes/directions (e.g., Simpson's paradox).
- Poor predictions.

Today's Robustness Question

What assumption/abstraction are we perturbing?

- That the nature of the relationship between x and y regarding moderation is appropriately specified.

What tools exist to evaluate robustness?

- Interaction models
- Smooth interactions with GAMs
- `interflex` family of tests
- Specification curves across multiple moderators

Today's Robustness Question

What assumption/abstraction are we perturbing?

- That the nature of the relationship between x and y regarding moderation is appropriately specified.

What are the limits of these tools?

- Cannot search for moderators - a moderator must be proposed by the researcher.
- Evidence for moderation is often low power.
 - Interaction effects are typically estimated much less precisely than average effects.
 - Reliable estimation requires sufficient support across combinations of x and the moderator.

Effect Heterogeneity

Consider the following linear model:

$$y_i = b_0 + b_1x_i + b_2z_i + e_i$$

In this model, the effect of x is:

$$\frac{\partial y}{\partial x} = b_1$$

The marginal effect is constant.

- In the previous lecture, we evaluated whether this assumption was reasonable across the values of x .
 - Is the effect of a one-unit change in x always the same regardless of the starting value?
- Here, we ask a related question:
 - Is the effect of x constant across the values of *other variables*?

Mediation vs Moderation

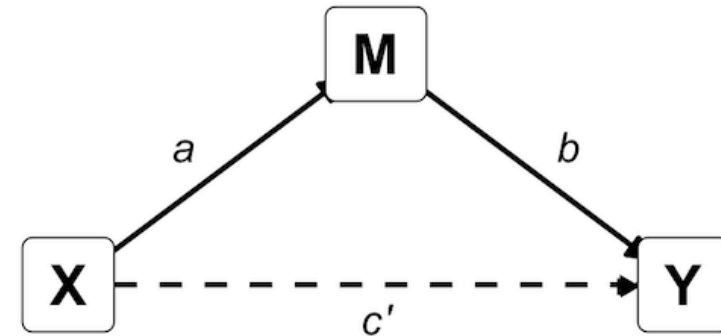
Moderation asks - does the direct effect of X vary by some other variable, W ?

- This is the effect heterogeneity of interest today.

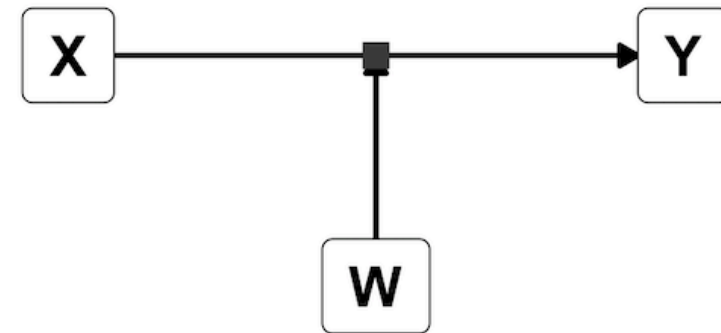
Mediation asks - is the effect of X direct and/or indirect through some other variable, M ?

- This is a related, but quite different question that we will not discuss today.

Mediation



Moderation



Moderation Effect

Suppose we are estimating a relationship between y and x and we suspect/hypothesize that relationship is moderated by z .

- We are really saying:

$$\frac{\partial y}{\partial x} = f(z)$$

The simplest $f(z)$ would just be linear:

$$\frac{\partial y}{\partial x} = b_1 + b_2 z_i$$

What model would generate this marginal effect:

$$y_i = b_0 + b_1 x_i + b_2 x_i z_i + b_3 z_i + \mathbf{g} \mathbf{w}_i + e_i$$

The marginal effect of x from the above equation is $b_1 + b_2 z_i$.

Robustness Workflow for Moderation

1. Estimate the target (unconditional) model.
2. Estimate a plausible moderated alternative.
3. Evaluate evidence for moderation.
4. Characterize the variation in effects.
5. Assess whether substantive conclusions change.

Typology of Models

Model	Assumption Change
Single interaction	Single effect varies across groups
Full interaction	All effects vary across groups
Subgroup Analyses	All effects + residual variances vary across groups

Interaction Effects

We often denote product regressors $x_i z_i$ as *interaction effects*.

Ultimately we want to know two things:

- Is there statistical evidence of a moderated effect?
- How variable is the moderated effect?
- Does the variation change or qualify the substantive conclusion?

These last two questions get to the heart of the robustness discussion.

Below, we walk through all the possible two-way interaction scenarios and we will discuss how to answer these questions.

Two Categorical Variables

With two categorical variables, essentially you are estimating a different conditional mean for every pair of values across the two categorical variables. You could do that as follows:

```
library(DAMisc)
library(car)
data(Duncan)
Duncan <- Duncan %>%
  mutate(
    inc.cat = cut(Duncan$income, 3),
    inc.cat = factor(
      as.numeric(inc.cat),
      labels=c("Low", "Middle", "High")))
mod <- lm(prestige~ inc.cat * type +
  education, data=Duncan)
```

```
library(broom)
tidy(mod)
```

```
## # A tibble: 10 × 5
##   term                estimate std.error statistic    p.value
##   <chr>              <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)         7.88      3.44      2.29  0.0279
## 2 inc.catMiddle       22.5      4.88      4.60  0.0000530
## 3 inc.catHigh        51.3      9.44      5.44  0.00000429
## 4 typeprof          55.6     11.7      4.76  0.0000330
## 5 typewc             2.54      8.12      0.314 0.756
## 6 education           0.280     0.112     2.50  0.0174
## 7 inc.catMiddle:typeprof -41.6     11.2     -3.70  0.000740
## 8 inc.catHigh:typeprof  -50.4     13.4     -3.76  0.000621
## 9 inc.catMiddle:typewc  -13.0     10.3     -1.26  0.215
## 10 inc.catHigh:typewc   -33.6     13.1     -2.56  0.0148
```

Evidence of Moderation

We evaluate evidence of moderation by evaluating the moderated model against the unconditional model:

```
uncond_mod <- lm(prestige~ inc.cat + type +
                 education, data=Duncan)
```

In this case, there is evidence of a moderated effect - `inc.cat`'s effect is moderated by `type` (and vice-versa).

```
anova(uncond_mod, mod, test="F")
```

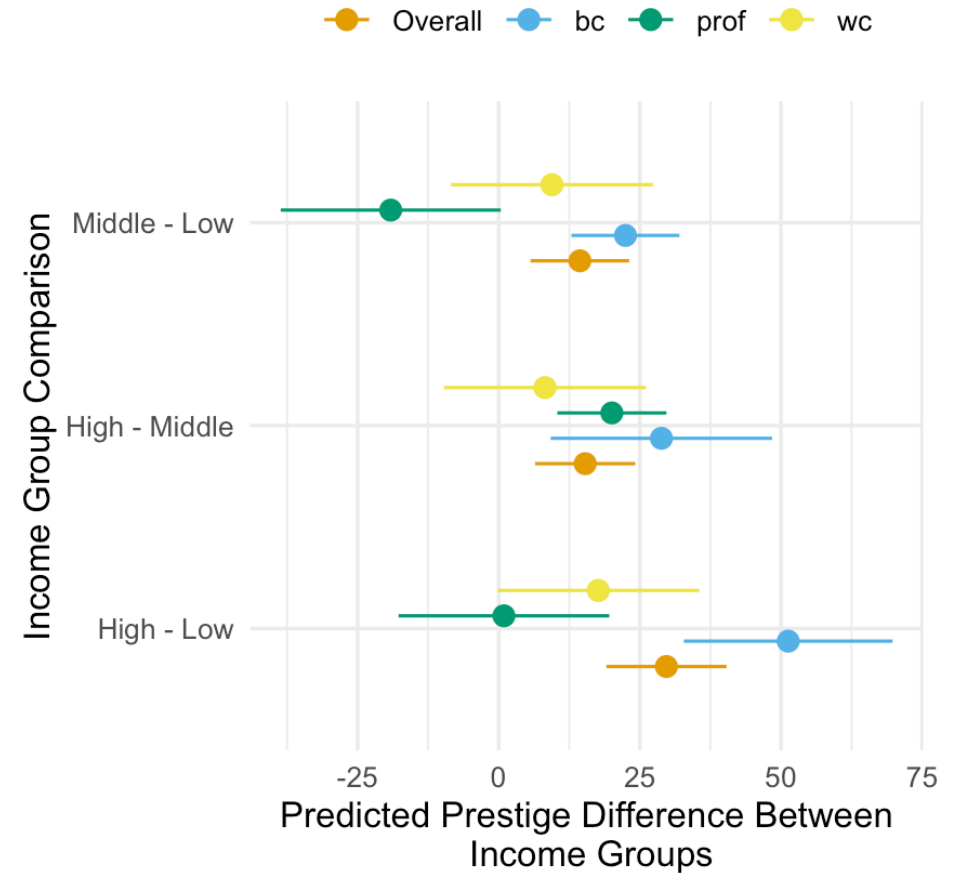
```
## Analysis of Variance Table
##
## Model 1: prestige ~ inc.cat + type + education
## Model 2: prestige ~ inc.cat * type + education
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1      39 4552.2
## 2      35 2907.7  4    1644.4 4.9484 0.002871 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The `anova()` function works because the unconditional model is nested in the conditional model.

Variation in Effects

There is considerable variation around the original effect.

- The main conclusions appear to not hold up - some positive effects are negative in some groups.





Robustness Evaluation

The unconditional relationship is not robust to this moderated alternative.

```
## # A tibble: 9 × 6
##   contrast      type    ovl    np    js    kl
##   <chr>        <fct> <dbl> <dbl> <dbl> <dbl>
## 1 High - Low   bc     0.14  0.123 0.738 2.83
## 2 High - Middle bc     0.315 0.309 0.477 1.31
## 3 Middle - Low bc     0.387 0.554 0.397 1.38
## 4 High - Low   prof   0.052 0.028 0.895 4.79
## 5 High - Middle prof   0.616 0.797 0.162 0.466
## 6 Middle - Low prof   0.018 0.006 0.958 6.08
## 7 High - Low   wc     0.386 0.432 0.392 1.07
## 8 High - Middle wc     0.525 0.536 0.256 0.629
## 9 Middle - Low wc     0.591 0.595 0.2    0.482
```

Two Categorical Summary

With these two categorical variables, the unconditional relationship fails to account for substantial heterogeneity across occupation types.

- The sensitive seems to vary by occupation type - WC sensitivity is low, but Professional sensitivity is quite high.
- Also, the difference between High and Low income seems to be the most sensitive contrast with generally lower overlap coefficients within each occupation type.

One Categorical + One Numeric

With one categorical variable, d (we'll use a binary variable here) and one numeric variable, x , we would have

$$y_i = b_0 + b_1x_i + b_2d_i + b_3x_id_i + e_i$$

We can think of the moderation effect in two ways:

Marginal Effect of x

$$\frac{\partial y}{\partial x} = b_1 + b_3d$$

Marginal effect of d

$$\frac{\partial y}{\partial d} = b_2 + b_3x$$

Interaction of Education and Sex

```
data(SLID, package="carData")
mod2_0 <- lm(log(wages) ~ education + sex, data=SLID)
mod2_1 <- lm(log(wages) ~ education * sex, data=SLID)
tidy(mod2_1)
```

```
## # A tibble: 4 × 5
##   term                estimate std.error statistic    p.value
##   <chr>              <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)        1.73      0.0492     35.3 1.69e-237
## 2 education          0.0577    0.00358    16.1 1.14e- 56
## 3 sexMale            0.477     0.0669      7.13 1.15e- 12
## 4 education:sexMale -0.0186   0.00489    -3.81 1.41e-  4
```

There is evidence of a moderation effect (i.e., a plausible alternative to the unconditional relationship).

Robustness and Effects of Education

Effect

```
## # A tibble: 5 × 5
##   model      sex      estimate std.error  p.value
##   <chr>    <chr>    <dbl>    <dbl>    <dbl>
## 1 Unconditional Female    0.0477  0.00244 4.65e-85
## 2 Unconditional Male      0.0477  0.00244 4.66e-85
## 3 Conditional  Female    0.0577  0.00358 1.95e-58
## 4 Conditional  Male      0.0391  0.00333 7.28e-32
## 5 Difference in Conditional (Male) - (Female) -0.0186  0.00489 1.39e- 4
```

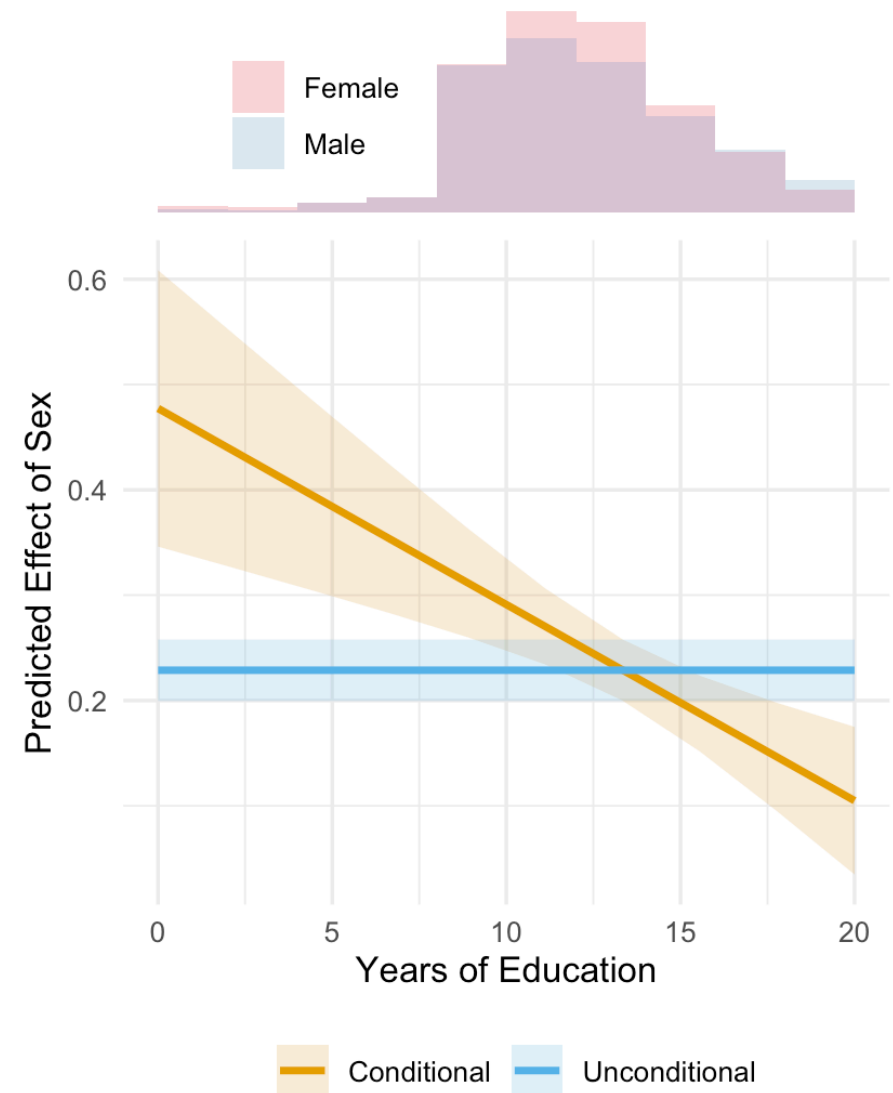
Robustness

```
## # A tibble: 2 × 5
##   sex      ovl      np      js      kl
##   <chr>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 Female 0.0949 0.0729 0.820  4.01
## 2 Male  0.133 0.124 0.758  3.44
```

Effect of Sex

The unconditional effect is flat and the conditional effect is negatively sloped.

- Note the density of education - it marks the regions where comparisons are permissible.
- The effects are always positive (males are predicted to earn more than females regardless of education).
- The difference between salaries of females and males decreases with education.



Robustness of Sex Effect

```
## # A tibble: 10 × 5
##   education    ovl      np      js      kl
##   <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1      0  0.00187 0.000502 0.994    7.93
## 2    2.22 0.00298 0.000776 0.991    7.62
## 3    4.44 0.00551 0.00146 0.985    7.19
## 4    6.67 0.0128 0.00391 0.970    6.49
## 5    8.89 0.0418 0.0203 0.914    5.20
## 6   11.1 0.209 0.247 0.639    2.60
## 7   13.3 0.993 0.950 0.0000609 0.000169
## 8   15.6 0.213 0.256 0.632    2.56
## 9   17.8 0.0426 0.0209 0.913    5.17
## 10  20 0.0130 0.00398 0.970    6.47
```

The sex effect is not robust as a constant effect, but the conclusion that men earn more remains robust over the observed range.

One Categorical + One Numeric Summary

Even though the robustness statistics are low, the substantive story is not overturned by moderation.

- The effects have the same signs, though their magnitudes differ.

Two Numeric Variables

With two numeric variables the interpretation gets a bit trickier. For example, consider the following model:

$$\hat{Y}_i = b_0 + b_1X_{i1} + b_2X_{i2} + b_3X_{i3} + b_4X_{i1}X_{i2}$$

We want to know the partial conditional effect of both X_1 and X_2 , but unlike above, neither can be boiled down to a small set of values. Just think about the equation:

$$\begin{aligned}\frac{\partial \hat{Y}}{\partial X_1} &= \beta_1 + \beta_4 X_2 \\ \frac{\partial \hat{Y}}{\partial X_2} &= \beta_2 + \beta_4 X_1\end{aligned}$$

Note, that β_4 is the amount by which the *effect* of X_1 goes up for every additional unit of X_2 and the amount by which the *effect* of X_2 goes up for every additional unit of X_1 .

Testable Hypotheses

$$\hat{Y}_i = b_0 + b_1X_{i1} + b_2X_{i2} + b_3X_{i3} + b_4X_{i1}X_{i2}$$

Berry, Golder and Milton (2012) suggest that we should be able to test 5 hypotheses:

- $\mathbf{P}_{X_1|X_2=\min}$ The marginal effect of X_1 is [positive, zero, negative] when X_2 takes its lowest value.
- $\mathbf{P}_{X_1|X_2=\max}$ The marginal effect of X_1 is [positive, zero, negative] when X_2 takes its highest value.
- $\mathbf{P}_{X_2|X_1=\min}$ The marginal effect of X_2 is [positive, zero, negative] when X_1 takes its lowest value.
- $\mathbf{P}_{X_2|X_1=\max}$ The marginal effect of X_2 is [positive, zero, negative] when X_1 takes its highest value.
- $\mathbf{P}_{X_1X_2}$ The marginal effect of each of X_1 and X_2 is [positively, negatively] related to the other variable.

Example

```
data(Prestige, package="carData")
Prestige$income <- Prestige$income/1000
addmod <- lm(prestige ~ income + education + type, data=Prestige)
intmod <- lm(prestige ~ income*education + type, data=Prestige)
tidy(intmod)
```

```
## # A tibble: 6 × 5
##   term                estimate std.error statistic    p.value
##   <chr>              <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)      -17.8      7.59     -2.34 0.0212
## 2 income             3.79      0.945      4.01 0.000124
## 3 education          5.10      0.777      6.57 0.00000000293
## 4 typeprof           5.48      3.71      1.48 0.144
## 5 typewc            -3.58      2.43     -1.48 0.143
## 6 income:education -0.210     0.0698     -3.01 0.00335
```


Is Moderation a Plausible Alternative?

- The `income:education` line answers this question. If it is significant, then there is a significant interaction, otherwise there is not.
- This is counter to a minor, though still influential, point in Brambor, Clark and Golder (2006), but is consistent with Berry, Golder and Milton (2012).
- The interaction provides evidence that moderation is a plausible alternative specification, so we evaluate robustness to it.



Robustness

```
# Income Robustness
add_slopes_inc <- slopes(addmod,
  newdata=datagrid(education =
    seq_range(Prestige$education, 10)),
  variables="income", by="education")
int_slopes_inc <- slopes(intmod,
  newdata=datagrid(education =
    seq_range(Prestige$education, 10)),
  variables="income", by="education")
rob_inc <-map_df(1:10, \(i)
  robustness(dnorm, dnorm,
    target_args = list(
      mean = add_slopes_inc$estimate[i],
      sd = add_slopes_inc$std.error[i]),
    alt_args = list(
      mean = int_slopes_inc$estimate[i],
      sd = int_slopes_inc$std.error[i]))) |>
  mutate(education = add_slopes_inc$education,
    .before="ovl")
```

```
# Education Robustness
add_slopes_ed <- slopes(addmod,
  newdata=datagrid(income =
    seq_range(Prestige$income, 10)),
  variables="education", by="income")
int_slopes_ed <- slopes(intmod,
  newdata=datagrid(income =
    seq_range(Prestige$income, 10)),
  variables="education", by="income")
rob_ed<- map_df(1:10, \(i)
  robustness(dnorm, dnorm,
    target_args = list(
      mean = add_slopes_ed$estimate[i],
      sd = add_slopes_ed$std.error[i]),
    alt_args = list(
      mean = int_slopes_ed$estimate[i],
      sd = int_slopes_ed$std.error[i]))) |>
  mutate(income = add_slopes_ed$income,
    .before="ovl")
```



Robustness Scores

Income Coefficients

rob_inc

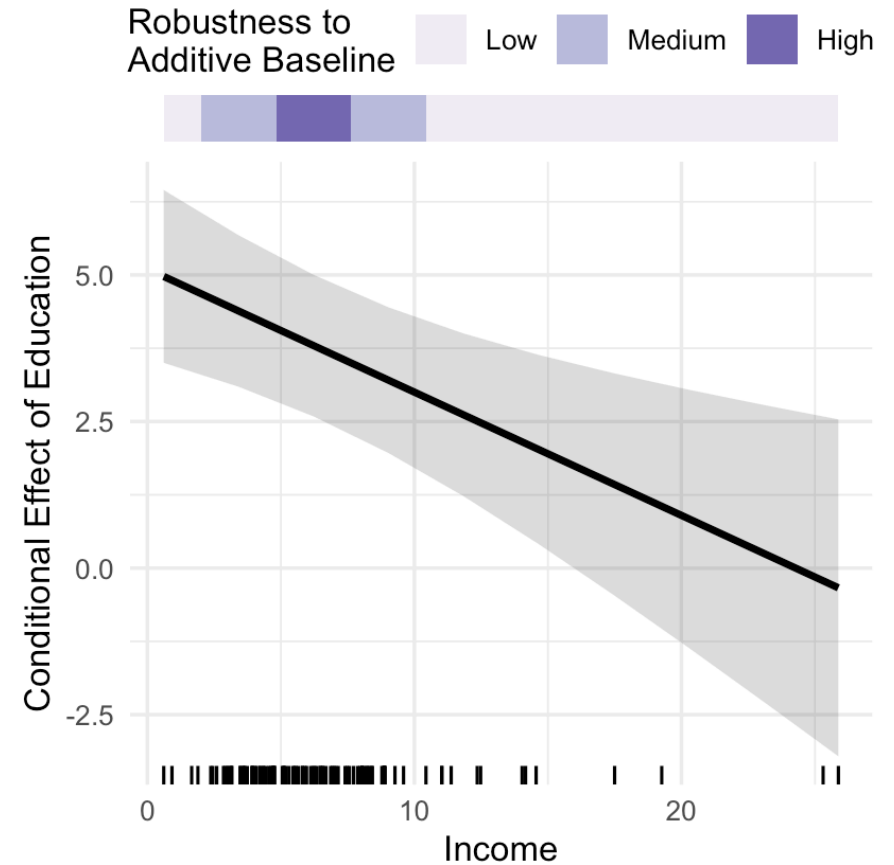
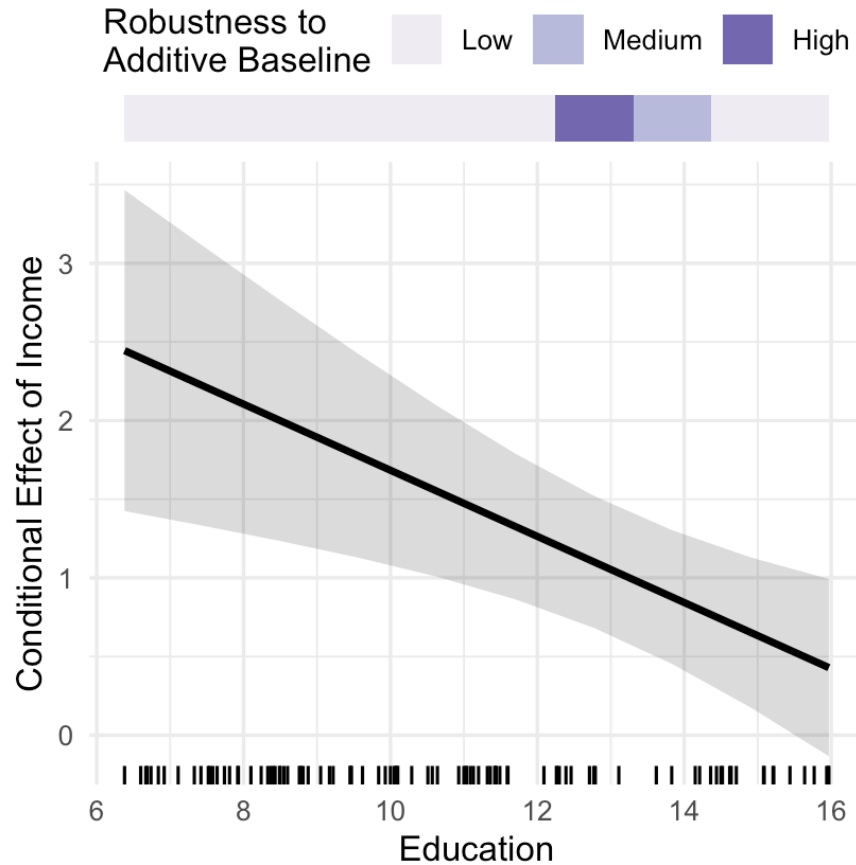
## # A tibble: 10 × 5					
##	education	ovl	np	js	kl
##	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	6.38	0.0484	0.0273	0.895	4.23
## 2	7.45	0.0682	0.0436	0.860	3.88
## 3	8.51	0.102	0.0786	0.803	3.42
## 4	9.58	0.163	0.160	0.705	2.78
## 5	10.6	0.279	0.355	0.536	1.92
## 6	11.7	0.494	0.695	0.275	0.879
## 7	12.8	0.839	0.939	0.0292	0.0855
## 8	13.8	0.756	0.910	0.0666	0.197
## 9	14.9	0.437	0.617	0.337	1.10
## 10	16.0	0.247	0.299	0.580	2.12

Education Coefficient

rob_ed

## # A tibble: 10 × 5					
##	income	ovl	np	js	kl
##	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	0.611	0.346	0.470	0.447	1.53
## 2	3.42	0.582	0.790	0.191	0.587
## 3	6.23	0.921	0.954	0.00736	0.0214
## 4	9.03	0.714	0.890	0.0911	0.272
## 5	11.8	0.432	0.609	0.343	1.13
## 6	14.6	0.259	0.320	0.564	2.06
## 7	17.5	0.160	0.156	0.710	2.82
## 8	20.3	0.104	0.0804	0.800	3.40
## 9	23.1	0.0716	0.0467	0.854	3.83
## 10	25.9	0.0520	0.0300	0.888	4.16

Conditional Effect + Robustness



Interpretation

- The effect of income is nearly always significant, though it gets smaller as education gets bigger. That is, as education increases, we expect smaller increases in prestige from increasing income
- The effect of education is significant and positive until around 16,000 dollars, which is around 2/3 the range of `income`, but is the 96th percentile because of the skewness of income.
- This suggests that people tend to derive prestige from either higher incomes or higher education, but not really both.



Berry, Golder and Milton Test

```
BGMtest(intmod, vars = c("income", "education"))
```

```
##               est      se      t p-value
## P(X|Zmin)  2.445 0.520  4.698 0.000
## P(X|Zmax)  0.429 0.287  1.495 0.138
## P(Z|Xmin)  4.756 0.712  6.681 0.000
## P(Z|Xmax) -0.335 1.466 -0.229 0.820
## P(XZ)      -0.210 0.070 -3.012 0.003
```

When Confidence Bounds Equal Zero

You may want to know when the confidence bounds are equal to zero. Consider the equation:

$$\hat{Y}_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \beta_4 X_{i1} X_{i2}$$

- We know that the conditional effect of X_1 is $\beta_1 + \beta_4 X_2$ and that the lower bound is $(\beta_1 + \beta_4 X_2) - 1.96 \times SE(\beta_1 + \beta_4 X_2)$.
- Since those are all quantities that we know (or estimate), we could set it equal to zero and solve.
- This is what the `changeSig` function does.



Change in Significance

```
changeSig(intmod, c("income", "education"))
```

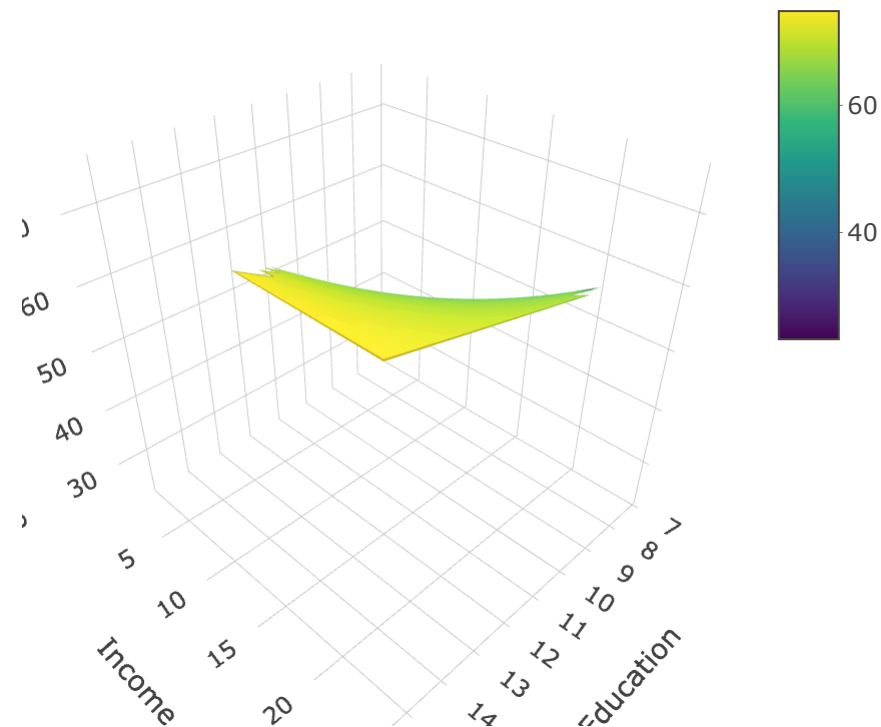
```
## LB for B(income | education) = 0 when education=15.4979 (95th pctlile)  
## UB for B(income | education) = 0 when education=27.9396 (> Maximum Value in Data)  
## LB for B(education | income) = 0 when income=15.9273 (96th pctlile)  
## UB for B(education | income) = 0 when income=59.5175 (> Maximum Value in Data)
```


3-D Surface

```
inc_seq <- seq_range(Prestige$income, 25)
ed_seq <- seq_range(Prestige$education, 25)

pred_3d <- function(model, variables, ...){
  d <- margineffects::datagrid(model = model,
                               grid_type = "mean_or_mode")
  o <- outer(variables[[1]], variables[[2]],
             Vectorize(function(x,y){
               d[[names(variables)[1]]] <- x
               d[[names(variables)[2]]] <- y
               predict(model, newdata=d)
             })))
  md <- insight::get_data(model)
  ch <- geometry::convhulln(md[,names(variables)])
  inch <- outer(variables[[1]], variables[[2]],
                Vectorize(function(x,y)geometry::inhulln(ch,
  o_in <- o
  o_in[which(!inch, arr.ind=TRUE)] <- NA
  res <- list(variables[[1]], variables[[2]], predictions=t(o),
names(res)[1:2] <- names(variables)
return(res)
}

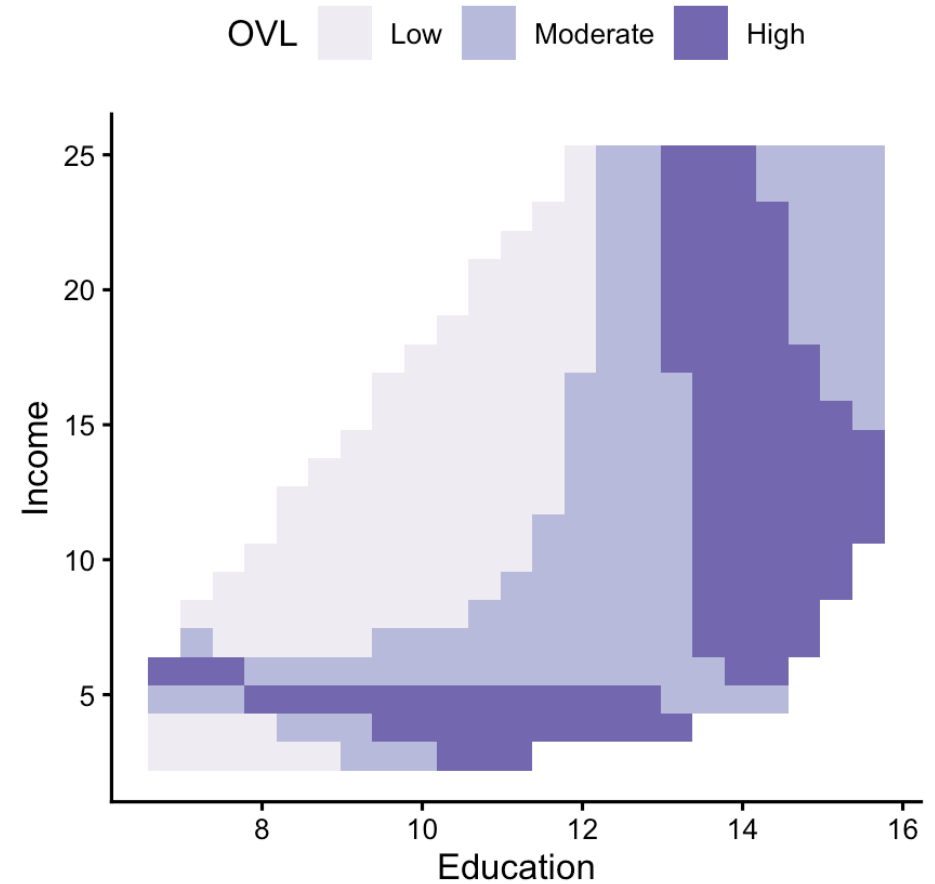
preds <- pred_3d(intmod,
  variables = list(education = ed_seq,
                   income = inc_seq))
```



Robustness of Predictions

The colored tiles are present for values in the convex hull.

- There are small, disconnected regions of high robustness.





Full Interaction Model

The fully interactive model would interact all variables with a grouping variable:

```
library(broom)
data(Prestige, package="carData")
Prestige$income <- Prestige$income/1000

fi_mod <- lm(prestige ~ (income + education)*type,
             data=Prestige)
slopes(fi_mod,
       by="type") |>
  filter_out(term == "type") |>
  as_tibble() |>
  select(type, term, estimate, std.error)
```

```
## # A tibble: 6 × 4
##   type term      estimate std.error
##   <fct> <chr>      <dbl>    <dbl>
## 1 bc    education  1.71     0.957
## 2 prof  education  3.10     0.863
## 3 wc    education  6.00     1.47
## 4 bc    income     3.52     0.556
## 5 prof  income     0.620    0.222
## 6 wc    income     1.45     0.700
```



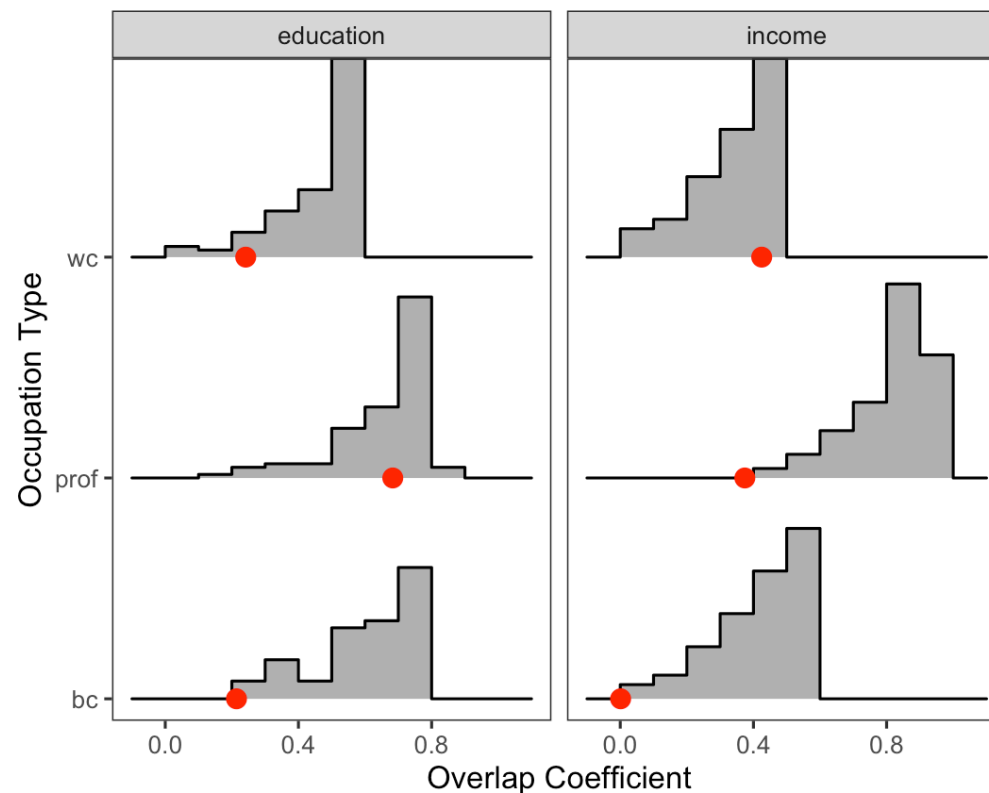
Full Interaction Robustness

```
## # A tibble: 6 × 7
## # Rowwise:
##   type  term      estimate std.error base_est base_se    ovl
##   <fct> <chr>      <dbl>      <dbl>    <dbl>    <dbl>  <dbl>
## 1 bc    education  1.71      0.957     3.67    0.640  0.214
## 2 prof  education  3.10      0.863     3.67    0.640  0.683
## 3 wc    education  6.00      1.47      3.67    0.641  0.242
## 4 bc    income    3.52      0.556     1.01    0.221  0.00112
## 5 prof  income    0.620     0.222     1.01    0.221  0.374
## 6 wc    income    1.45      0.700     1.01    0.221  0.425
```

Simulation

Most of the coefficients are quite sensitive to the no moderation assumption.

- Income for White Collar and Education for Professional incomes appear to be about what we would expect if the additive model is true.
- The other effects have overlap coefficients significantly lower than we would expect if the additive model were, in fact, true.





Subgroup Analysis

Sometimes, people prefer estimating subset models - it makes interpretations a bit easier immediately.

Subgroup analysis relaxes other assumptions:

- all effects vary by group.
- residual variances differ.

Subgroup Robustness

```
## # A tibble: 6 × 6
## # Rowwise:
##   type      n term      estimate base_est    ovl
##   <fct> <int> <chr>      <dbl>    <dbl>  <dbl>
## 1 prof     31 income    0.620     1.24  0.157
## 2 prof     31 education 3.10      4.29  0.275
## 3 bc       44 income    3.52      1.24  0.00168
## 4 bc       44 education 1.71      4.29  0.0307
## 5 wc       23 income    1.45      1.24  0.435
## 6 wc       23 education 6.00      4.29  0.231
```

Robustness and Effect Heterogeneity

Approach	Robustness Question
Single interaction	Does a single effect vary?
Full interaction	Do all effects vary?
Subgroup Analyses	Do the relationships survive in distinct populations?